

Erdős Problem #778

EP-778

L1 - TRACTABLE

PARTIALLY SOLVED

Graph Theory

ORIGINAL PROBLEM STATEMENT

Alice and Bob play a game on the edges of K_n , alternating colouring edges by red (Alice) and blue (Bob). Alice goes first, and wins if at the end the largest red clique is larger than any of the blue cliques. Does Bob have a winning strategy for $n \geq 3$? (Erdős believed the answer is yes.)

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Archive: <https://www.unsolvedmath.com/problems?difficulty=1>
Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>